

E3974152_1

The equation $x^4 + x^2 + xy + y^3 + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$x^4 + x^2 + xy + y^3 + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = -H_2$ and $d_2 = 2H_1$, and the product formula for the Hilbert symbol of one pair of values. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x^2 + xy + y^3 + 3$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= xy + 2x + y^2 - y - 4, \\ T_1 &= x^6 - x^5 - x^4y + 12x^4 + x^3y^2 + x^3y - 9x^3 + 2x^2y^2 \\ &\quad - 14x^2y + 18x^2 + 6xy^2 + 31xy + 9x + y^2 - 39y + 1, \\ U_1 &= -x^3y + 2x^3 + x^2y - 7xy - 4x + 6y + 10, \\ V_1 &= 4, \\ G_1 &= x^3y + 2x^3 + 2x^2y - 10x^2 + 6xy + 26x + y - 40, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= 2x - 2y - 1, \\ T_2 &= 8x^4 + 8x^3y + 8x^2y^2 - 4x^2y - 8x^2 - 10xy - 18x - 10y^2 - 4y + 11, \\ U_2 &= -4x^3 - 2x^2 + 2x + 1, \\ V_2 &= 3, \\ G_2 &= -16x^2 + 20, \end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\begin{aligned} \lambda_1 &= 1184x^3 - 1616x^2 - 984x - 4124, \\ \mu_1 &= 1184x^4 + 592x^3y - 3688x^3 - 1992x^2y + 3324x^2 - 1184xy^2 + 3196xy \\ &\quad - 3682x + 3096y^2 - 3610y + 3867, \\ \nu_1 &= -2368x + 6192, \end{aligned}$$

and, for the signs, we also use

$$\begin{aligned} Q_2 &= 4x^2 + 4xy + 4y^2 - 2y + 1, \\ M_2 &= 8x^4 + 8x^3 + 4x^2 + 2x + 23, \\ E_2 &= 48x^2 + 16x + 12. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 + H_2V_1^2 = G_1F, \tag{2}$$

$$H_2T_2 - U_2^2 - 2H_1V_2^2 = G_2F, \tag{3}$$

$$\lambda_1 H_1 + \mu_1 H_2 + \nu_1 F = 5 \cdot 79^2, \quad (4)$$

$$H_1 = (x + y - 3)(y + 2) + 2, \quad (5)$$

$$-8F = Q_2 H_2 - M_2, \quad (6)$$

$$16Q_2 = (4x + 8y - 2)^2 + E_2. \quad (7)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$A = -d_2 = -2H_1, \quad B = -d_1 = H_2, \quad T_A = -2T_1, \quad T_B = T_2.$$

By (2)–(3), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$AT_A = (2U_1)^2 - B(2V_1)^2, \quad BT_B = U_2^2 - AV_2^2. \quad (8)$$

Step 1: signs. *The sign of H_1 .* Suppose that $H_1 \leq 0$. Then (5) gives $(x + y - 3)(y + 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $x + y \geq 4$ and $y \leq -3$, or $x + y \leq 2$ and $y \geq -1$. In the first case, write $x + y = 4 + X$ and $y = -3 - Y$ with $X, Y \geq 0$, so that $x = X + Y + 7$ and $y = -Y - 3$. Then $F(x, y) = W_{1,1}(X, Y)$, where

$$\begin{aligned} W_{1,1}(X, Y) = & X^4 + 4X^3Y + 28X^3 + 6X^2Y^2 + 84X^2Y + 295X^2 + 4XY^3 + 84XY^2 \\ & + 589XY + 1383X + Y^4 + 27Y^3 + 285Y^2 + 1349Y + 2405. \end{aligned}$$

All coefficients of $W_{1,1}$ have the same sign and its constant term is non-zero, so $W_{1,1}(X, Y) \neq 0$. In the second case, write $x + y = 2 - X$ and $y = -1 + Y$ with $X, Y \geq 0$, so that $x = -X - Y + 3$ and $y = Y - 1$. Then $F(x, y) = W_{1,2}(X, Y)$, where

$$\begin{aligned} W_{1,2}(X, Y) = & X^4 + 4X^3Y - 12X^3 + 6X^2Y^2 - 36X^2Y + 55X^2 + 4XY^3 - 36XY^2 \\ & + 109XY - 113X + Y^4 - 11Y^3 + 51Y^2 - 107Y + 89. \end{aligned}$$

Expanding $W_{1,2}(3 + X, 3 + Y)$, and, for $0 \leq i < 3$, $W_{1,2}(i, 3 + Y)$ and $W_{1,2}(3 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $W_{1,2}(i, j) \neq 0$ for $0 \leq i, j < 3$. Hence $W_{1,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (6) gives $Q_2 H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_2 > 0$ and $E_2 > 0$. Hence $M_2 > 0$ and $E_2 > 0$, so (7) gives $16Q_2 \geq E_2 > 0$, and $Q_2 H_2 = M_2 > 0$ gives $H_2 > 0$.

By these signs, $A < 0$ and $B > 0$.

Hence Theorem 2(i) gives $(A, B)_\infty = 1$ (as $B > 0$).

Step 2: primes outside S . Let $S = \{2, 3, 5, 79\}$ be the set of primes dividing 2, the constants 1 and 2 in the $d_k = cH_{k'}$, $V_1 = 4$ and $V_2 = 3$, and the right-hand side of (4). Let $p \notin S$ be an odd prime; we show that $(A, B)_p = 1$. If $p \nmid AB$, this follows from Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid A$, then $p \mid H_1$, so $p \nmid H_2$ by (4) and $p \nmid B$; as $p \nmid 2V_1$, Lemma 3 with $(a, b, r, c, t) = (A, B, 2U_1, 2V_1, T_A)$ and (8) gives $(A, B)_p = 1$; if $p \mid B$, then $p \mid H_2$, so $p \nmid H_1$ by (4) and $p \nmid A$; as $p \nmid V_2$, Lemma 3 with $(a, b, r, c, t) = (B, A, U_2, V_2, T_B)$ and (8) gives $(B, A)_p = 1$, which equals $(A, B)_p$ by Theorem 2(v).

Step 3: the primes in S . By Step 1, A and B are non-zero. If $T_A \neq 0$, then (8) and Lemma 4 give $(A, B)_v = (T_A, B)_v$, and if $T_B \neq 0$, they give $(A, B)_v = (B, A)_v = (T_B, A)_v = (A, T_B)_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 16^2 residue pairs modulo 16, we find that (1) has exactly 8 solutions modulo 16, and that at each of them one of the cases listed for $p = 2$ in Table 1 holds. Each case gives the stated value of $(A, B)_2$. Hence $(A, B)_2 = -1$.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 2 solutions modulo 3, and that at each of them one of the cases listed for $p = 3$ in Table 1 holds. Each case gives the stated value of $(A, B)_3$. Hence $(A, B)_3 = 1$.

The prime 5. Checking the 5^2 residue pairs modulo 5, we find that (1) has exactly 9 solutions modulo 5, and that at each of them one of the cases listed for $p = 5$ in Table 1 holds. Each case gives the stated value of $(A, B)_5$. Hence $(A, B)_5 = 1$.

The prime 79. Checking the 79^2 residue pairs modulo 79, we find that (1) has exactly 102 solutions modulo 79, and that at each of them one of the cases listed for $p = 79$ in Table 1 holds. Each case gives the stated value of $(A, B)_{79}$. Hence $(A, B)_{79} = 1$.

Step 4: contradiction. By Theorem 2(iv), $(A, B)_\infty \prod_p (A, B)_p = 1$, the product over all primes, and by Step 2, the factors with $p \notin S$ are 1. Hence $\prod_{p \in S} (A, B)_p = (A, B)_\infty = 1$. But by Step 3, $\prod_{p \in S} (A, B)_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A, B)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 16				
1	(T_A, B)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 5$	-1
2	(T_A, B)	$2^{2i+1} \cdot 3$	$2^{2i} \cdot 5$	-1
3	(T_A, B)	$2^{2i+1} \cdot 5$	$2^{2i} \cdot 5$	-1
4	(T_A, B)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 5$	-1
$p = 3$, solutions modulo 3				
1	$3 \nmid AB$			1
2	(A, T_B)	$3^{2i}(-)$	$3^{2i}(+)$	1
$p = 5$, solutions modulo 5				
1	$5 \nmid AB$			1
2	(A, B)	$5^{2i}(+)$	any	1
3	(A, T_B)	$5^{2i}(+)$	$5^{2i}(+)$	1
4	(T_A, B)	$5^{2i}(+)$	any	1
$p = 79$, solutions modulo 79				
1	$79 \nmid AB$			1
2	(A, T_B)	$79^{2i}(+)$	$79^{2i}(-)$	1

	symbol	first entry	second entry	value
3	(A, T_B)	$79^{2i}(+)$	$79^{2i}(+)$	1
4	(T_A, B)	$79^{2i}(+)$	any	1